CHAPTER 3

Vascular Assessment and Imaging Techniques for Critical Limb Ischemia in Patients with Diabetes Mellitus

Neal R. Barshes

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INTRODUCTION

The arterial system delivers oxygen and nutrients to tissue. The arterial circulation of the lower extremity is relevant to those clinicians who manage foot wounds because some minimum level of arterial perfusion is needed for tissue viability and for wound healing. The actual delivery of nutrients to peripheral tissues is difficult to measure in vivo, but the oxygen consumption of oxygen can be estimated by three components: (a) amount of blood delivered per unit time (milliliter per minute [mL/min]), (b) oxygen content of blood (milliliter oxygen per liter of blood), and (c) the fraction of oxygen extracted from delivered blood (percentage).

A typical cardiac output for an average-sized man is approximately 5 L/min. The brain receives approximately 10% to 15% of the cardiac output, and the kidneys receive about 20% to 25%. The mean blood flow through the popliteal artery in young, healthy patients at rest is 55 mL/min (22), or approximately 1% of the cardiac output. The amount of oxygen bound to hemoglobin is more than 6,000 higher than that found free in the plasma. As seen in Table 3.1, hemoglobin levels have a relatively higher impact on oxygen content than oxygen saturation. The oxygen extraction fraction of the lower extremity at rest is generally 30% to 45% and may approach 90% or more with activity (23,45). Ischemic limbs can be seen to have lower oxygen partial pressures and lower pH in mixed venous blood than limbs that have undergone revascularization (15). The oxygen consumption of tissues in the foot is much lower than that of other, more active tissue. Myocardium consumes 8 mL O_2/min per 100 mm^3 tissue at rest and 77 mL O_2/min per 100 mm^3 tissue when active. The kidney consumes 5 mL O_2/min per 100 mm^3 tissue. In comparison, gastrocnemius and soleus muscles at rest consume approximately 0.5 mL O_2/min per 100 mm^3 tissue, and skin consumes 0.2 mL O_2/min per 100 mm^3 tissue (25).

WOUND HEALING AND ARTERIAL PERFUSION: THEORETICAL CONCERNS

Wound healing requires anabolic metabolism. Perfusion probably needs to be higher in the setting of wound healing than at steady state. Whole-body caloric requirements increase 20% with mild stress and may increase 60% or more with severe stress or infection (12). It is not known how much regional blood flow varies (increases) with wound healing. Oxygen tension decreases to 50% of normal with bone healing and to 35% of normal with bone healing and infection (2).

Poiseuille law (Fig. 3.1) describes the impact of various variables on flow in the context of nonpulsatile laminar flow. Radius is the major determinant because of the exponential relationship it has with flow; length has an impact as well. When major arteries are occluded, blood flow takes "collateral" pathways to the periphery. As a rule, these collaterals are smaller caliber and increased length (and therefore higher resistance) compared to the primary artery, and this is the reason why blood flow preferentially travels through collateral pathways only when the primary artery is significantly narrowed or occluded. Although the collateral vessels present are numerous, they generally do not provide for as much flow as the primary artery because of their relatively decreased caliber and increased length (Fig. 3.2). Even if making the liberal assumption of collaterals that are 40% of the diameter of the main artery and 30% longer, 51 collaterals of these dimensions would be required to provide the same degree of flow as the primary artery.

Pulsatile flow makes things much more complex, and Poiseuille law may greatly underestimate energy losses in pulsatile systems. Turbulence increases energy losses even further still. All this is to say that although two- or three-dimensional imaging of atherosclerotic lesions can identify peripheral artery disease and perhaps give some signal of its severity, it cannot quantify its hemodynamic impact.

TABLE 3.1 Oxygen Content (in mL Oxygen per L Blood) at Various Levels of Hemoglobin, Oxygen Saturation, and Plasma Oxygen Partial Pressures

	Hemoglobin 15	Hemoglobin 12	Hemoglobin 10	Hemoglobin 8
SpO_2 = 99%, PaO_2 = 145 torr	20.65	16.61	13.91	11.22
SpO_2 = 97%, PaO_2 = 96 torr	20.09	16.13	13.49	10.85
SpO_2 = 95%, PaO_2 = 79 torr	19.62	15.75	13.16	10.58
SpO_2 = 92%, PaO_2 = 65 torr	18.97	15.22	12.71	10.21
SpO_2 = 88%, PaO_2 = 55 torr	18.12	14.53	12.14	9.74

•AQ6• PaO_2, ; SpO_2, .

Normal Arterial Anatomy

Arterial blood ejected from the left ventricle toward the lower extremities passes through the aorta to the lower abdomen and pelvis. The aorta bifurcates into the left and right common iliac arteries, generally at the level of the fourth lumbar vertebra. This vessel becomes the external iliac artery distal to the origin of the internal iliac (also known as hypogastric) artery and the common femoral artery after passing under the inguinal ligament. The common femoral artery is at the level of the femoral head; it can also be found through palpation caudal to a line connecting the anterior superior iliac spine of the ilium and the pubic tubercle of the pubis bone but cephalad to the inguinal crease of the proximal thigh.

The common femoral artery bifurcates into the deep femoral artery (also known as the *profunda femoris* artery)—a lateral branch that distributes throughout the musculature of the thigh—and the superficial femoral artery. The superficial femoral artery spirals through the thigh, from its anterior position in the groin, and then medial in the thigh after passing under through a hiatus (opening) in the adductor magnus muscle and becoming the popliteal artery,

$$Q = \frac{\pi \cdot R^4 \cdot (P_1 - P_2)}{8\eta L}$$

FIGURE 3.1 Poiseuille law. η, coefficient of viscosity (expressed •AQ2• in dynes*sec*cm^2); L, length; P, pressure; R, radius.

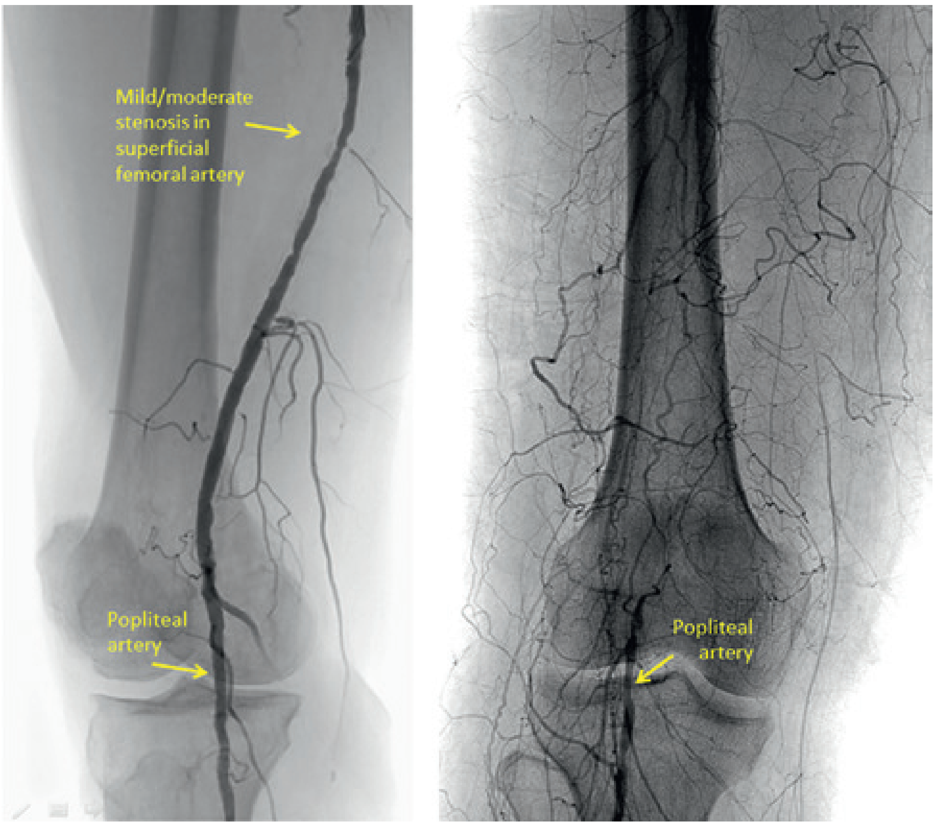


FIGURE 3.2 Angiogram demonstrating blood flow through a normal-caliber superficial femoral and popliteal artery *(left panel)* and through collaterals in the setting of an occluded superficial femoral and proximal popliteal artery *(right panel)*.

to a position posterior in the leg at the level of the knee. Most people have two pairs of bifurcations at the level of the proximal tibia. At the first bifurcation, the popliteal artery divides into the anterior tibial artery and a segment referred to as the tibioperoneal trunk. The tibioperoneal trunk then bifurcates into the posterior tibial and peroneal (fibular) arteries.

The posterior tibial artery divides into the medial and plantar arteries distal to malleoli at the ankle. The anterior tibial artery becomes the dorsalis pedis artery after passing under the extensor retinaculum of the distal ankle. The lateral plantar and dorsalis pedis arteries generally coalesce to form the deep (primary) plantar arch, but many additional communications among the tibial arteries exist (4).

Arterial blood passes through these named arteries and their branches and into arterioles, terminal arterioles, and then capillaries before returning to the venous circulation initially via postcapillary venules. It is these components of the arterial tree that are referred to as the "microcirculation" (<300 μm in diameter) and that are involved in nutrient and gas exchange in soft tissues. Blood supply to the skin is supplied by three types of arterioles: direct cutaneous vessels, musculocutaneous perforators, and the fasciocutaneous system (11).

Relationship between Diabetes Mellitus and Peripheral Arterial Disease

Diabetes mellitus affects the microvasculature of the lower extremity. Atherosclerotic lesions in patients with diabetes mellitus are generally found in the distal superficial femoral artery, the popliteal artery, and the tibial arteries (18,37) (Fig. 3.3). Many patients presenting with nonhealing ulcers or other tissue loss will be found to have isolated tibial disease (21). Lesions of the aortoiliac (suprainguinal) circulation are generally more often associated with significant smoking history (35). At least one pedal vessel is patent and suitable for distal revascularization in the foot in 70% of patients with diabetes mellitus (versus 80% of patients without diabetes mellitus) and only 50% in those with renal insufficiency with or without diabetes mellitus (13). Sparing of vessels in the foot allows for revascularization options in the majority of patients (34).

Primary arterial calcification, including Mönckeberg sclerosis and internal elastic lamina calcification (20), refers to calcification of the media. This finding is present in approximately 60% of patients presenting with acute diabetes mellitus–associated foot problems (1). It has also been associated with chronic kidney disease and other systemic diseases (20). There is some relationship with peripheral arterial disease (PAD) because PAD was present in 60% of those with medial calcification (versus 30% of those without this finding, odds ratio of 4) (1).

The concept of "small vessel disease" continues to cause confusion among physicians. Diabetes mellitus does alter the microvasculature. First recognized in the 1980s, these changes include (a) thickening of the basement membrane, (b) abnormal arteriovenous shunting (8,29), (c) decreased capillary density, and (d) an increased incidence tunica media smooth muscle hyperplasia causing occlusion of dermal arterioles (average of 12% in one histologic study) (19), and can result in slower wound healing.

The Identification of Peripheral Arterial Disease Using History and Physical Exam Findings

The clinician should ask about a history of previous revascularization or revascularization attempts (e.g., previous stents in leg arteries, previous leg bypass operations) in either the index or contralateral lower extremity. Unfortunately, little other history and few symptoms have very reliable association with PAD. A history of claudication symptoms does have some association with PAD overall, but most patients presenting with diabetic foot ulcers do not have such a history (perhaps because of the presence of sensory neuropathy or the distribution of atherosclerotic disease in patients with diabetes mellitus) (9,10).

Palpation of peripheral pulses—specifically, the femoral, popliteal, posterior tibial, and dorsalis pedis pulses—constitute the most important exam component when identifying PAD.

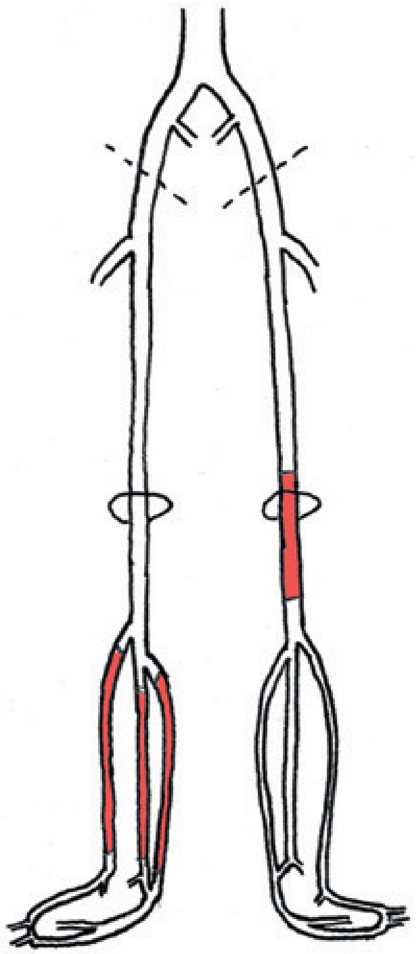


FIGURE 3.3 Anatomic areas in which atherosclerotic lesions commonly occur among patients with diabetes mellitus and peripheral arterial disease.

Palpating a pulse is a simple physical exam maneuver, but results should be considered with some degree of caution. This caution is based on the fact that the interrater reliability of a peripheral pulse exam in most studies is lacking quality. Agreement on the presence of peripheral pulses also seemed higher when attempting to categorize as normal, reduced, or absent rather than simply present.

Many clinicians evaluate pedal pulses using a handheld (continuous-wave) Doppler probe. Finding *both* dorsalis pedis *and* posterior tibial artery Doppler signals to be absent or monophasic may signify severe PAD, but the presence of biphasic signals over either or both arteries does *not* rule out hemodynamically significant PAD.

Although many clinicians believe hair loss, atrophic skin, or cool temperature skin are signs of PAD, findings from the Seattle Diabetic Foot Study suggest their diagnostic accuracy is very poor (likelihood ratio of 1.5 when present, 0.81 when absent) (9). The presence of any lower extremity bruit—iliac, femoral, or popliteal location—may be helpful in ruling in PAD, as the absence of a bruit in these three locations may decrease the likelihood of PAD (27).

Dependent rubor is a term describing redness present in the periphery of the foot (generally toes and forefoot) when the limb of patients changes to pallor when elevated above the level of the heart. The rubor results from maximal dilation of the microvasculature of the foot in the context of more proximal occlusions. The pallor occurs because the change in elevation further limits marginal perfusion to the foot by eliminating the contribution hydrostatic pressure has in providing arterial perfusion to the limb. Dependent rubor is generally a considered a relatively specific sign of severe PAD, although no studies have evaluated its diagnostic accuracy.

QUANTITATIVE ARTERIAL NONINVASIVE TESTING: GENERAL CONCEPTS

The theoretical relationship between arterial perfusion to the foot (at least as measured by some of the means described later) and wound healing has been derived empirically by a few studies, and it is thought to be sigmoidal in shape (24). In short, wound healing does not require normal or near-normal levels of arterial perfusion. What we do not know is (a) if there is some minimum level versus variation among patients and wounds and (b) how to best quantify arterial perfusion.

Some have proposed that the concept of "angiosomes" is relevant to the evaluation and management of arterial perfusion of the foot. In general terms, an angiosome is defined as the three-dimensional volume of tissue that primarily receives blood supply from a particular artery. Attinger and colleague's (4) cadaver studies have identified five angiosomes in the foot: those of the dorsalis pedis artery, the lateral plantar artery, the medial plantar artery, the calcaneal branch of the peroneal artery, and the calcaneal branch of the posterior tibial artery. Several studies have demonstrated that direct revascularization of the artery associated with the angiosome where a foot ulcer is located (a so-called "direct" revascularization) is associated with higher limb preservation rates than revascularization of an artery that is not associated with the angiosome in which the ulcer is located (a so-called "indirect" revascularization). The concept of angiosomes, however, has been investigated primarily through injection studies in cadavers. The concept assumes that all the main arteries in the foot are end arteries without complementary areas of perfusion. Other studies have suggested that territories of arterial perfusion are much more variable in vivo and that the difference between the outcomes of direct versus indirect revascularization is eliminated if patency of the pedal arch is considered. Nonetheless, the idea that the vascular evaluation of the foot should not be only a global assessment of arterial perfusion but rather specific to the area in which the foot ulcer is located remains important.

Ankle Pressures and the Ankle-Brachial Index

Ankle pressures are the absolute systolic pressure of blood flow detected in the posterior tibial artery or the anterior tibial artery/dorsalis pedis artery. The ankle-brachial index (ABI) is defined as the ratio of the highest ankle pressure to the highest arm pressure. The highest arm pressure is taken as the best representation of central (aortic) pressure because, unlike ankle pressures, arm pressure measurements are rarely falsely elevated by vessel noncompressibility. An interarm systolic blood pressure differential of at least 10 mm Hg was present in 9.4% of the Framingham Heart Study population (mean age of 61 years, free of cardiovascular disease at baseline [43], and 26% of a population undergoing vascular surgery operations [14]).

There are two methods of obtaining ABI measurements. At the most basic, a manual blood pressure cuff is placed at the level of the ankle. A handheld (continuous-wave) Doppler probe is used to auscultate an arterial signal over the posterior tibial artery at the level of the malleolus. The cuff is manually inflated at least 20 mm Hg past the pressure required to extinguish the Doppler signal; 250 mm Hg or more pressure may be needed. The cuff is then slowly deflated—rate of 3 to 4 mm Hg per second. The pressure at which the Doppler signal over the artery reappears is the ankle pressure for the posterior tibial artery. The process is repeated by holding the Doppler probe over the dorsalis pedis artery in the dorsum of the foot. Blood pressures are obtained in both arms with the pressure detected by either auscultation over the brachial artery (i.e., listening for Korotkoff sounds) near the antecubital fossa or Doppler probe auscultation over the radial artery at the wrist (42). It is worthwhile noting that results represent the systolic pressure at which phasic arterial flow returns at the level of the blood pressure cuff, *not* the level of the Doppler probe.

An appropriately sized cuff is 20% wider than the diameter of the limb. The cuff should go circumferentially around the limb (Fig. 3.4A). Aneroid blood pressure cuffs (i.e., those with a clock face measuring pressure in the cuff at any given time)

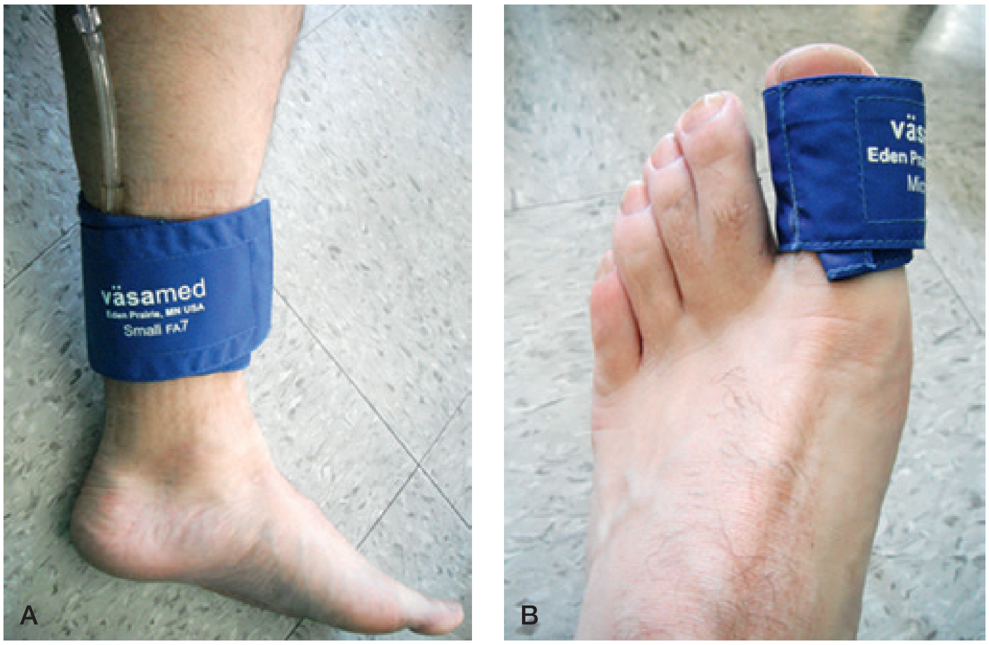


FIGURE 3.4 Ankle (A) and toe (B) pressure measurements using appropriately sized cuffs.

are very inexpensive and very portable. These features and the fact that they are used for the measurement of arm pressures should make the measurement of ABIs available to clinicians in all but the most severely resource-limited health care environment. Most noninvasive vascular laboratories obtain pressures using automated cuffs. As with aneroid (manual) cuffs, these automated cuffs can be the same as those used for the measurement of blood pressure in the arm. These automated devices use oscillations in air pressure within the bladder of the cuff to detect the ankle pressure (or arm pressure). PAD is defined as an ABI ratio less than or equal to 0.9. In the setting of nonhealing ulcers, absolute ankle pressures of less than 70 mm Hg are indicative of severely impaired arterial perfusion (32).

As mentioned earlier, ankle pressures and ABIs can be quickly obtained with very inexpensive equipment generally available in most health care settings. The specificity (probability that a normal test result correctly rules out significant PAD) is excellent at 89% (6). There are many important limitations in the use of ankle pressures and ABIs in the setting of diabetic foot wounds. It is widely recognized that falsely elevated pressures (generally ABI >1.3) represent noncompressible vessels and therefore yield no relevant diagnostic information. Less frequently appreciated, however, is the fact that results below this level, including values considered normal or near-normal, can also represent falsely elevated results that erroneously suggest to the clinician that significant PAD is absent. Overall, sensitivity of ABIs is 61% (6) and is lower than that of other diagnostic tests to be discussed later. Use of the *lower* ankle pressure in generating an ankle-brachial ratio may actually improve sensitivity in identifying significant PAD (16) but is not widely done.

Finally, similar pressure measurements at higher levels in the limb (often high thigh, low thigh, and calf) are also often obtained when ankle pressures are obtained in noninvasive vascular laboratories. A decrease of >10 to 20 mm Hg between any two consecutive levels may help identify the level of a hemodynamically significant atherosclerotic lesion and therefore may help in case planning prior to revascularization attempts. Still, it is the measurement of ankle pressures and ABIs (or other measures described later) that provides direct information on the overall perfusion to the foot itself.

Toe Pressures

Given that calcification of tibial (including ankle-level) arteries is so common among patients with diabetes mellitus, the measurement of toe (digit) pressures developed as a method of measuring blood flow that would not be affected by this process. Measurements are taken with the patient in supine position. Digit cuffs are used on either the first or second toe (see Fig. 3.4B); other toes are too small for measurement. As with ankle pressures and ABIs, these can be obtained manually but are most often performed using an automated plethysmography cuff. Whereas automated plethysmography cuffs used for ankle pressures often use air, toe pressure cuffs uses photoplethysmography. Toe cuffs emit an infrared light, and the amount of this infrared light reflected by the tissue corresponds to pulsatile blood flow and tissue volume.

A toe-brachial index (TBI) ratio of greater than 0.7 or an absolute toe pressure greater than 60 mm Hg should be considered indicative of adequate arterial perfusion to the foot. Toe pressures less than 30 mm Hg are generally interpreted as indicating severe PAD, but risk of limb loss may be high only with pressures even lower than this (30). The toe pressure should be interpreted as being the most sensitive indicator of PAD in cases when a discrepancy between toe pressure and ankle pressure exists (e.g., normal ankle pressures, severely impaired toe pressures) (31).

TABLE 3.2	Overall Sensitivity and Specificity of Individual Tests Included among the Diagnostic Testing Strategies			
Variable	Sensitivity	Specificity	PPV	NPV
PE	53.3 (52.1–54.6)	82.6 (82.2–83.1)	42.5 (41.4–43.7)	88.0 (87.6–88.4)
ABI	61.0 (59.7–62.1)	89.1 (88.6–89.6)	74.7 (73.7–75.9)	81.2 (80.6–81.8)
SPP	81.7 (79.9–83.6)	79.3 (77.2–81.1)	81.0 (78.9–83.1)	80.1 (78.2–82.4)
$TcPo_2$	83.0 (81.8–84.3)	62.8 (61.2–64.4)	66.7 (65.7–68.1)	80.6 (79.1–82.2)
TBI	84.0 (82.8–85.0)	77.8 (76.1–79.5)	86.7 (85.8–87.8)	73.7 (71.9–75.2)

•AQ7• NPV, negative predictive value; PE, pulse exam; PPV, positive predictive value.

Discrepancies are generally due to false-normal ankle pressures (due to medial calcification) rather than pedal-level atherosclerotic disease (i.e., "small vessel disease").

Toe pressures appear to have excellent sensitivity and specificity (Table 3.2) (6). Measurement is quick and not known to be affected by ambient temperature or other conditions. The first or second toe must be present and without a wound that would preclude measurement, and therefore, toe pressure measurements may not be possible to obtain in all individuals.

Skin Perfusion Pressures

Skin perfusion pressures (SPPs) attempt to directly measure the pressure of skin perfusion in various regions of the foot. These are obtained using a laser Doppler sensor applied to the skin in the area of interest. A cuff is placed around this sensor and inflated. A curve showing the degree of perfusion is generated as the cuff is slowly deflated (Fig. 3.5). The SPP is defined as the point at which skin perfusion increases above the level seen with full inflation. Measurement during slow deflation induces reactive hyperemia, and this is thought to prompt a more sensitive assessment of perfusion.

Similar to toe pressures, SPP measurements less than 30 mm Hg should be interpreted as indicating severe PAD (41). SPP measurements between 30 and 60 mm Hg suggest mild or moderate impairment, and those over 60 mm Hg suggest no significant arterial impairment is present.

Pooled analysis of previously published studies (see Table 3.2) (6) suggests that SPPs have excellent sensitivity and specificity in the detection of significant PAD. The equipment

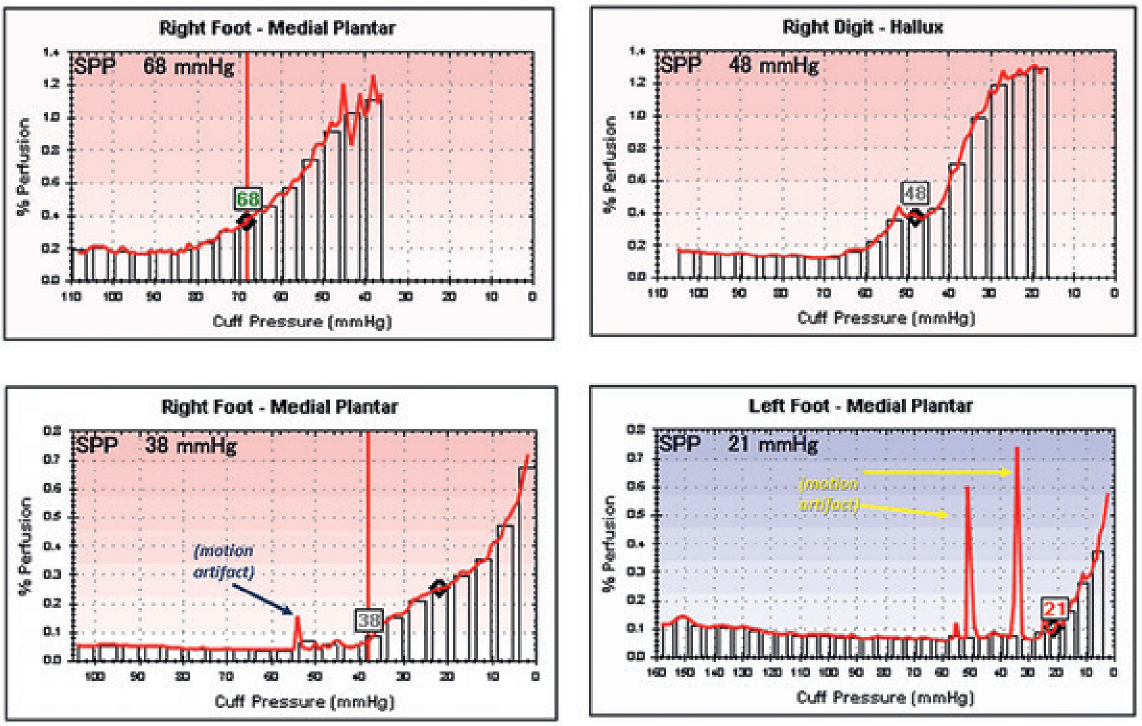


FIGURE 3.5 Skin perfusion pressure (SPP) measurements: normal *(top left)*, mild impairment *(top right)*, moderate impairment *(bottom left)*, and severe impairment *(bottom right)*.

needed for testing is compact and mobile, and obtaining and interpreting requires only limited training. These features make SPPs appealing for point-of-care testing. Unlike toe pressures, SPPs can be obtained anywhere an appropriately sized cuff can be fit. Therefore, measurements specific to the area/angiosome of interest can be obtained, including the lateral forefoot, the midfoot, and the heel. Measurement is done via the laser Doppler sensor and is therefore thought to be somewhat less susceptible to vessel calcification.

Equipment used to measure SPPs is not as widely available as that needed for TBIs. Measurement may be affected by motion artifacts that occur frequently with involuntary movements of the foot (see Fig. 3.5, *lower two panels*). Measurement takes 1 to 2 minutes, and pressure applied by the cuff during this duration may be uncomfortable to some patients. Like other modalities, the relationship between various SPP levels and wound healing is incompletely understood, but it is not clear if there are specific SPP thresholds below which revascularization would be strongly recommended.

Transcutaneous Oxygen Pressures

Transcutaneous oxygen pressure ($TcPo_2$) measurements are obtained using a small electrode that is typically placed on a location of the dorsal forefoot away from any bone, hair, hyperkeratotic tissue, or ulcer. Testing must be done with the patient in supine position and in a room with an ambient temperature of 21° to 23°C (70° to 74°F). Testing is often done after a 15- to 20-minute period of acclimatization to these conditions.

$TcPo_2$ measurements have been correlated to clinical outcomes in several studies (17,28,44). Specifically, $TcPo_2$ measurements less than 30 mm Hg are considered to indicate severely impaired arterial perfusion and low probability of wound healing. Measurements ranging between 30 and 50 mm Hg are considered to indicate moderately or mildly impaired arterial perfusion and a somewhat decreased probability of wound healing (17).

Measurements can be taken in any non-bony area of intact skin in the foot. $TcPo_2$ measurements can therefore quantify regional blood flow to specific areas of interest in the foot. As noted earlier, measurements require stable ambient conditions and 10 to 15 minutes for acclimatization. Measurements are then often done continuously for up to 30 minutes, so total testing time may require 45 minutes or more for $TcPo_2$ measurements alone. $TcPo_2$ measurements also appear to have less specificity than SPPs or toe pressures (63% versus 78% to 79%, respectively, in a pooled analysis; see Table 3.2) (6).

Arterial Blood Flow Velocities

Some noninvasive vascular laboratories obtain and report arterial blood flow velocities of major arteries at various levels throughout the lower extremity. Very elevated velocities (typically 250 cm/s or higher) are taken to represent areas of hemodynamically significant stenosis. Conversely, very low velocities (<40 cm/s) may also signal a hemodynamically significant stenosis or occlusion at a more proximal level.

At this time, the use of velocities to evaluate arterial perfusion of the foot is not conventional. Suggested thresholds for velocities representing hemodynamically significant lesions are not universally agreed upon, and there is less data evaluating the diagnostic accuracy of velocities than data evaluating other diagnostic modalities. Measuring native artery velocities throughout the limb is time-consuming, and results can be influenced by the experience and technique of the ultrasound technician more than other testing modalities. Certain arteries or arterial segments can also be difficult to visualize because of their location, especially the peroneal artery (because of its depth).

Adjunctive Vascular Testing Assessments

The use of pulse oximetry as a bedside screening test to identify PAD has previously been described. It can be used similar to ABIs, using the return of a pulse oximetry waveform as a substitute for Doppler signal. Similar to ABIs, this can be used to generate an index, referred to as the Lanarkshire oximetry index. These values appear to correlate with ABIs (7).

One group of investigators studied 84 patients. They considered abnormal any of the following results: (a) toe pulse oximetry 2% or more lower than finger oximetry and (b) a decrease of 2% or more when elevating the leg by 12 in. in height. The gold standard was "significant lower extremity PAD" as defined by the presence of monophasic Doppler signals. This group found that the presence of one or both criteria has an 86% sensitivity in identifying PAD and absence of both had a specificity of 92% (36). Another group of investigators examining 40 patients and 40 random volunteers found that all patients without PAD had normal toe oximetry readings but had only a 28% sensitivity in detecting moderate PAD (ABI 0.5 to 0.9) and 77% sensitivity in detecting severe PAD (ABI <0.5) (26).

Indocyanine green (ICG) imaging allows for real-time visualization of relative skin perfusion. Ratios of skin perfusion in various regions of interest can be calculated after some reference region, an area thought to have normal or sufficient skin perfusion, is identified. These measures therefore represent a relative value. Because the validity of these relative values is critically dependent on the reference region having normal or sufficient blood flow, this assessment should probably be paired with a more conventional test that at least assesses global foot perfusion. Additional quantitative methods of using the absolute rate of ICG influx and efflux have been described (40) as a means to circumvent this limitation, but these have not yet been well validated.

The quantitative assessment of skin temperature as a sign of PAD probably also has limited utility, as skin temperature varies significantly with ambient temperatures (39). Absolute temperatures by themselves probably have little role. Patients with PAD may display temperature decreases with exercise (24) and temperature increases following intervention (38) but mirror changes seen in ABIs, a much more standardized and well-accepted measurement.

Analysis of dorsal pedal venous blood gas (venous oximetry) has previously been described. Partial pressure of venous blood oxygen in patients included in this study increased after revascularization (15). No standardized values of venous oximetry for the foot or lower extremity have been evaluated, however, nor have levels or degrees of improvement that suggest an “adequate” revascularization. The clinical utility of venous oximetry in the diagnosis or quantification of lower extremity PAD is therefore limited at this time.

Qualitative Assessment

Plethysmography (pulse volume recordings) provides a graphical representation of total blood flow at various levels of the limb. Similar to the measurement of ankle pressures or other segmental pressures, plethysmography is obtained by placing an appropriately sized cuff at the level of interest in the limb. Instead of measuring the pressure at which a systolic waveform returns after temporary occlusion, plethysmography graphs the rate and volume of displacement of air in the cuff throughout the cardiac cycle. Plethysmography recordings at various levels are frequently obtained in noninvasive vascular labs and reported along with their corresponding pressure measurements.

Plethysmography is considered a secondary measurement of arterial perfusion because its interpretation is ultimately quantitative. In general, normal waveforms have a sharp (brief) systolic upstroke and feature the dicrotic notch (representing reversal of flow during early diastole; Fig. 3.6A). Mildly impaired waveforms do not have a dicrotic notch but have an obvious systolic upstroke and a downward slope that is either convex or neutral (see Fig. 3.6B). Moderately impaired plethysmography waveforms have a detectable systolic upstroke that is somewhat more rounded (longer duration) and a diastolic component that is concave (see Fig. 3.6C). Severely impaired waveforms show little or no obvious systolic upstroke.

Although an assessment with handheld (continuous-wave) Doppler for the global assessment of blood flow has important limits mentioned earlier, handheld Doppler can be used for a qualitative assessment of the patency of the deep plantar arch or other collaterals in the foot when a foot operation may disrupt collaterals. Such situations may include a proximal transmetatarsal or Lisfranc amputation that will disrupt the deep plantar arch. Attinger et al. (3,5) have shown more details on how a handheld Doppler may be used to assess the patency of the pedal arch, and Roukis (36) has demonstrated details on how a handheld Doppler assessment may be used in preoperative planning prior to creation of local soft tissue flaps in the foot.

Cross-sectional Imaging

In general, noninvasive anatomic imaging should not be used as first-line testing in diabetic patients with foot ulcers. Although imaging can identify the presence of PAD, it cannot quantify the hemodynamic effects. As such, the clinical impact of atherosclerotic lesions is often unclear and may predispose to either under treatment or over treatment. There are roles for noninvasive anatomic imaging—at least after the presence of significant PAD has been noted from noninvasive arterial tests. In health care environments with suboptimal access to angiographic suites for angiography and possible endovascular intervention, anatomic imaging may help elucidate the level(s) of atherosclerotic disease that will require intervention. Having this information may help in planning the optimal route of arterial access (e.g., retrograde via contralateral femoral versus antegrade via ipsilateral femoral). It may also help in planning efficient use of the angiography suite, as patients whose noninvasive anatomic imaging suggests surgical bypass may be the only feasible option can be scheduled for only as much time as is needed for a diagnostic angiogram, whereas those who seem to have endovascular options may need more time.

There are many limitations in the use of noninvasive anatomic imaging, especially in the setting of a diabetic foot ulcer. Computed tomographic arteriography (CTA; CT imaging •AQ3• with arterial-phase contrast) involves exposure to radiation and typically requires large doses (>100 mL) of iodinated contrast agents to image from the distal aorta to the feet. The spatial resolution of CTA and the presence of calcification of the media may impair the ability to accurately evaluate the patency of tibial and pedal arteries. Magnetic resonance angiography (MRA) uses a gadolinium, a noniodinated contrast agent that is safe to use in patients with an estimated glomerular filtration rate greater than 30 mL/min. MRA tends to overestimate the degree of stenosis. MRA cannot visualize

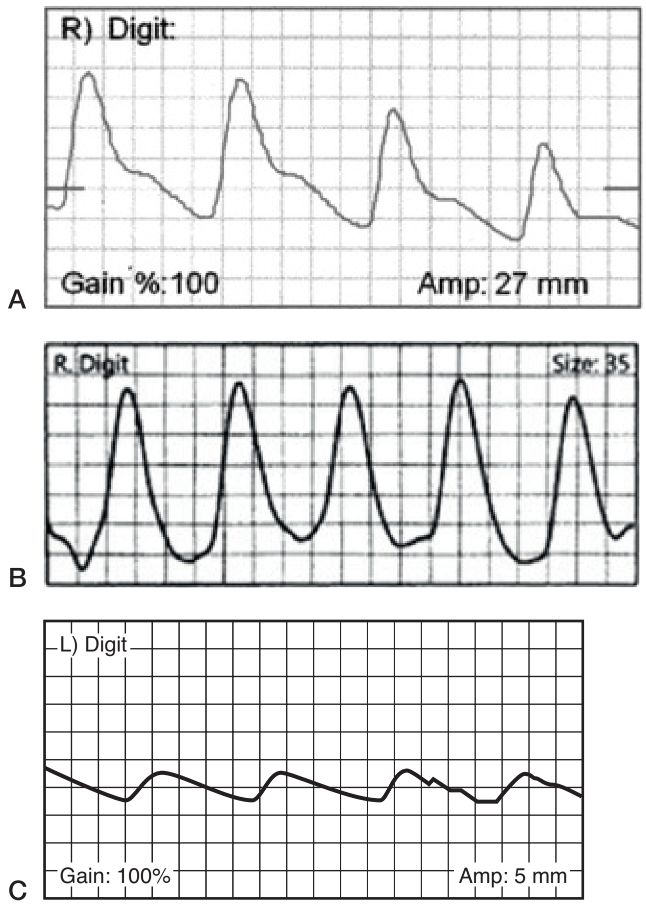


FIGURE 3.6 Plethysmography: normal **(A)**, mildly impaired **(B)**, and moderately impaired **(C)**.

areas with previously placed stents and thus cannot be used to identify occlusions or stenosis within segments previously treated by stents.

When to Order Noninvasive Studies

Angiography is still considered the gold standard for the identification of significant PAD. Not all patients with diabetic foot ulcers should be subjected to angiography, however, because of its associated costs and invasive nature, a screening test is therefore helpful in this situation. In general, ideal screening tests are widely available and are low cost. Although angiography is the confirmatory test having 100% specificity, it is more important that a screening test used to determine the need for angiography have high *sensitivity*—that is, can rule out significant PAD with high levels of accuracy.

Our group has utilized probabilistic decision analysis models to estimate the composite (overall) sensitivity and specificity of various combinations of tests used to identify PAD among patients with diabetes mellitus. Diagnostic accuracy of individual tests, including pulse exam, were pooled from previously published studies (see Table 3.2). These results were then included into a probabilistic Markov model to obtain estimates of various diagnostic strategies (i.e., the overall diagnostic accuracy of thirteen combinations of sequential diagnostic tests, including pulse exam). The median sensitivity, specificity, and median negative predictive values are presented in Table 3.3 (6).

The results of this analysis suggested that employing certain strategies may increase the identification of PAD among those with foot ulcers, perhaps while also decreasing unnecessary costs:

1. Toe pressures and SPPs are preferable to ankle pressures/ABIs.
2. The sensitivity of SPPs, TBIs, and $TcPo_2$ measurement are similar, but specificities differ. Therefore, SPPs may be the single best noninvasive testing modality. TBIs/toe pressures should be used when SPP measurement is not available. $TcPo_2$ measurement can be used when SPP and TBI measurement is not available.
3. Performing noninvasive testing on all patients and using this to determine the need for angiography (regardless of the results of pulse exam; strategies 8 to 10) yields good levels of sensitivity (see Table 3.3).
4. Excellent levels of sensitivity in the detection of PAD are obtained when noninvasive testing is used to corroborate the absence of PAD in patients with palpable pulses and when an angiography is performed in patients without palpable pulses (i.e., without using noninvasive testing to confirm PAD; strategies 11 to 13) (see Table 3.3).

It may also be appropriate to perform noninvasive arterial testing after revascularization. Performing such testing early after revascularization may help ensure the procedure has augmented arterial perfusion that will be sufficient for wound healing. Testing may be helpful in patients with a remote history of revascularization and recent development of a foot ulcer to determine the current patency of the revascularization procedure.

TABLE 3.3 Cumulative Sensitivity, Specificity, and Negative Predictive Value of Various Combinatorial Testing Strategies, Ordered by Increasing Sensitivity, in the Base Case Scenario (Population Peripheral Arterial Disease Prevalence of 9.8%) •AQ8•

Strategy No.	Brief Description	Median Sensitivity	Median Specificity	Median NPV
1	PE: if abnl, ABI; if abnl, DSA	32.6 (31.6–33.6)	97.4 (97.2–97.6)	69.9 (66.5–73.5)
2	PE: if abnl, SPP; if abnl, DSA	43.7 (42.4–45.1)	96.5 (96.0–96.8)	74.0 (70.0–76.6)
3	PE; if abnl, $TcPo_2$; if abnl, DSA	44.3 (43.0–45.7)	93.5 (93.2–93.9)	73.6 (70.0–76.2)
4	PE: if abnl, TBI; if abnl, DSA	44.8 (43.7–46.2)	96.1 (95.8–96.4)	74.3 (70.5–76.9)
5	PE: if abnl, DSA	53.3 (52.1–54.6)	82.6 (82.2–83.1)	74.7 (70.8–77.2)
6	ABI: if abnl, DSA	60.9 (59.9–62.1)	89.1 (88.6–90.0)	79.1 (75.7–81.2)
7	PE: if nl, ABI, if abnl, DSA	81.8 (81.1–82.5)	73.6 (73.0–74.2)	87.0 (84.7–88.6)
8	SPP: if abnl, DSA	82.0 (80.0–83.6)	89.1 (88.6–89.6)	89.1 (87.0–90.7)
9	$TcPo_2$: if abnl, DSA	83.1 (81.8–84.4)	62.8 (61.3–64.5)	86.1 (83.3–87.6)
10	TBI: if abnl, DSA	84.0 (83.0–85.2)	77.8 (76.1–79.4)	88.9 (86.9–90.3)
11	PE: if nl, SPP, if abnl, DSA	91.6 (90.7–92.4)	65.7 (63.7–67.2)	92.8 (91.2–93.9)
12	PE: if nl, $TcPo_2$, if abnl, DSA	92.1 (91.4–92.8)	51.9 (50.6–53.4)	91.7 (89.8–92.6)
13	PE: if nl, TBI, if abnl, DSA	92.6 (92.1–93.1)	64.2 (62.8–65.7)	93.4 (92.1–94.3)

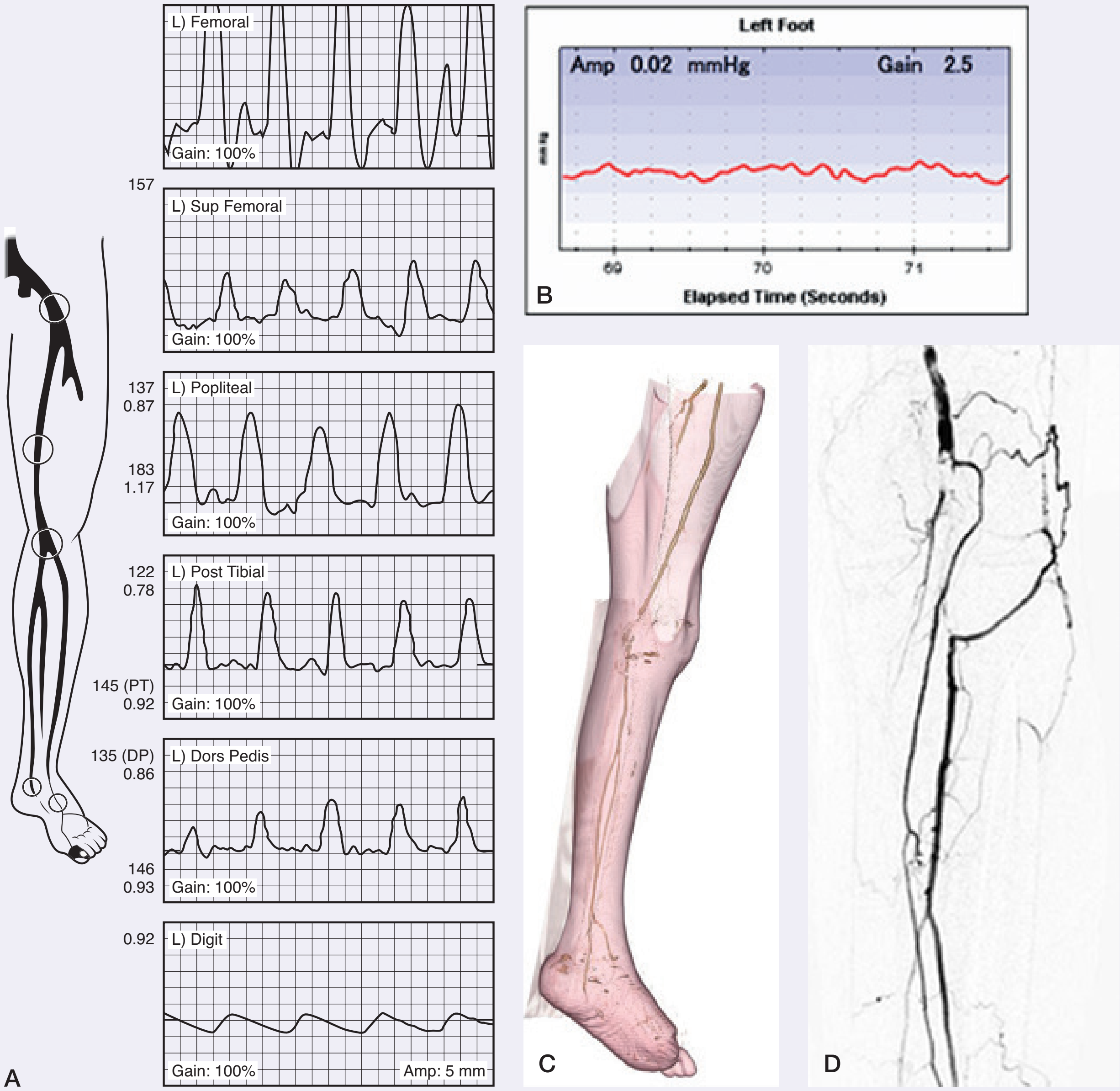
•AQ9• Values in parentheses represent the 25%–75% interquartile range of values.
abnl, abnormal; DSA, digital subtraction angiography; nl, normal; NPV, negative predictive value; PE, pulse exam.

CLINICAL CASE SCENARIOS

Four cases are presented to further demonstrate some of the principles described earlier. All cases had some form of anatomic imaging available for correlation. Three of the four cases had either more than one noninvasive testing modality done at the initial assessment or repeat assessment after revascularization. This frequency of testing is not always necessary and should not be interpreted as typical or recommended; rather, these cases are somewhat exceptional but are presented because comparing the results of the various noninvasive testing modalities helps demonstrate their respective strengths and weaknesses.

CASE STUDY 1

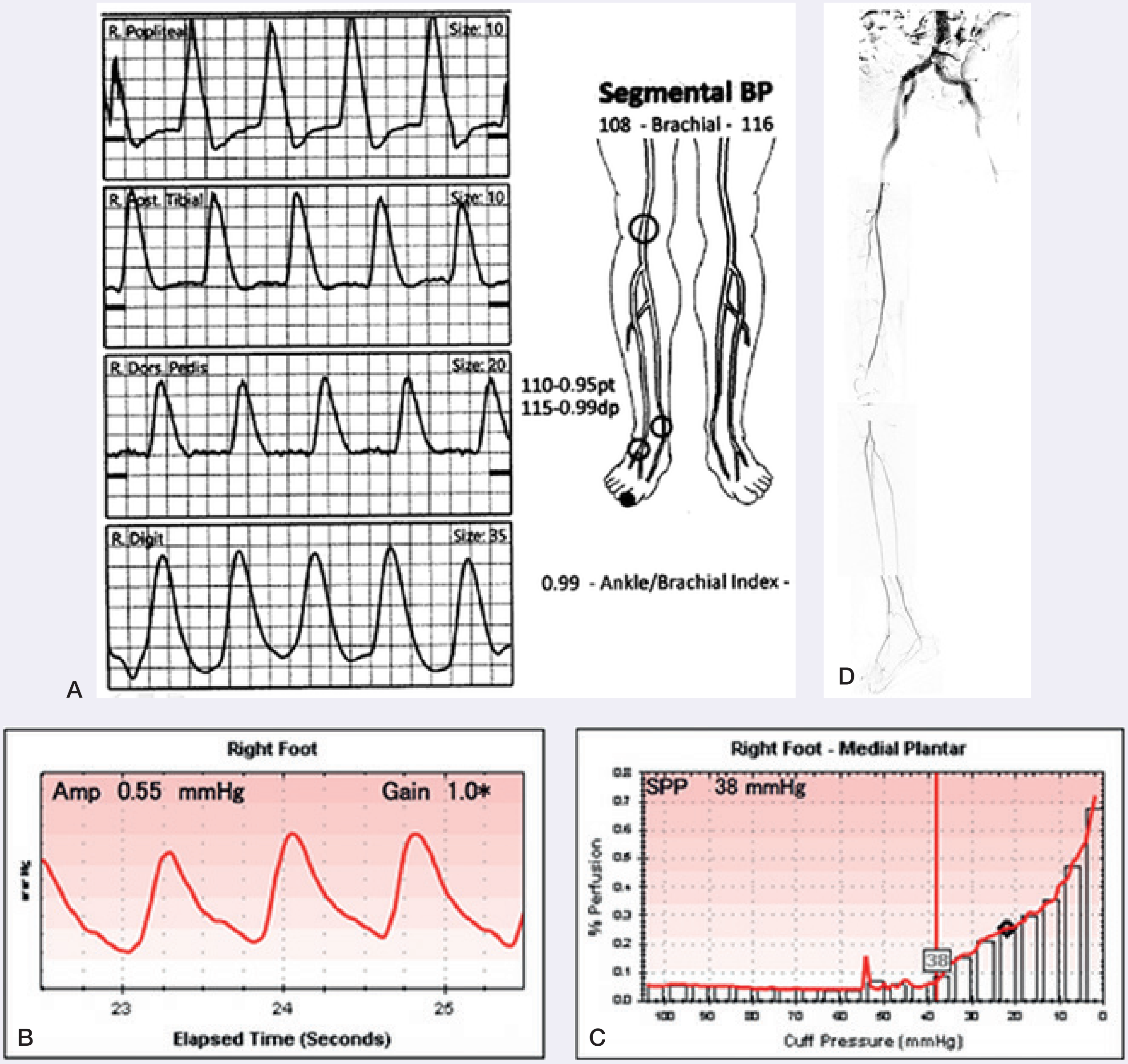
A 64-year-old man with diabetes mellitus (hemoglobin A_{1c} 5.7%) presented with gangrene of the tip of the left first toe and forefoot cellulitis. Transmetatarsal plethysmography suggested severe impairment (Fig. 3.7A,B). He underwent an initial open amputation of the first toe through the metatarsophalangeal joint to treat the infection, followed by a bypass from the distal thigh portion of the superficial femoral artery to the tibioperoneal trunk using ipsilateral nonreversed great saphenous vein and subsequent amputation of the distal metatarsal with primary wound closure (see Fig. 3.7C,D). The wound healed completely without recurrence of infection.



•AQ5• **FIGURE 3.7** Demonstration of an arterial bypass from the distal thigh portion of the superficial femoral artery to the tibioperoneal trunk using an ipsilateral nonreversed great saphenous vein.

CASE STUDY 2

A 65-year-old man with diabetes mellitus (hemoglobin A_{1c} 6.0%) and stage 3A chronic kidney disease (serum creatinine 1.8 mg/dL, estimated glomerular filtration rate of 46 mL/min) presented with a right second toe ulcer. He had a history of transtibial amputation of the left leg and partial right hallux amputation. The right femoral and popliteal pulses were palpable; pedal pulses were not palpable. Toe pressures could not be obtained because of swelling of the second digit and previous partial amputation of the hallux. The ABI was 0.99, and right fifth toe plethysmography suggested only mild impairment (Fig. 3.8A–C). Transmetatarsal plethysmography also showed mild impairment, but SPP obtained along the medial aspect of the plantar forefoot (under the head of the second metatarsal) yielded a pressure of 36 mm Hg. Because of somewhat discrepant results and the patient's previous history of contralateral leg amputation (i.e., at high risk for a significant change in functional status if amputation of the ipsilateral limb were to occur), angiography was performed. This demonstrated inline blood flow from the distal aorta to the right foot via patent tibial and pedal vessels that were without hemodynamically significant lesions (see Fig. 3.8D; angled views at the level of the knee demonstrated a patent popliteal artery—not seen in the anteroposterior projection shown here because of a metallic knee prosthesis). The patient underwent second ray (toe and metatarsal) resection. The surgical site healed completely without the need for arterial intervention.

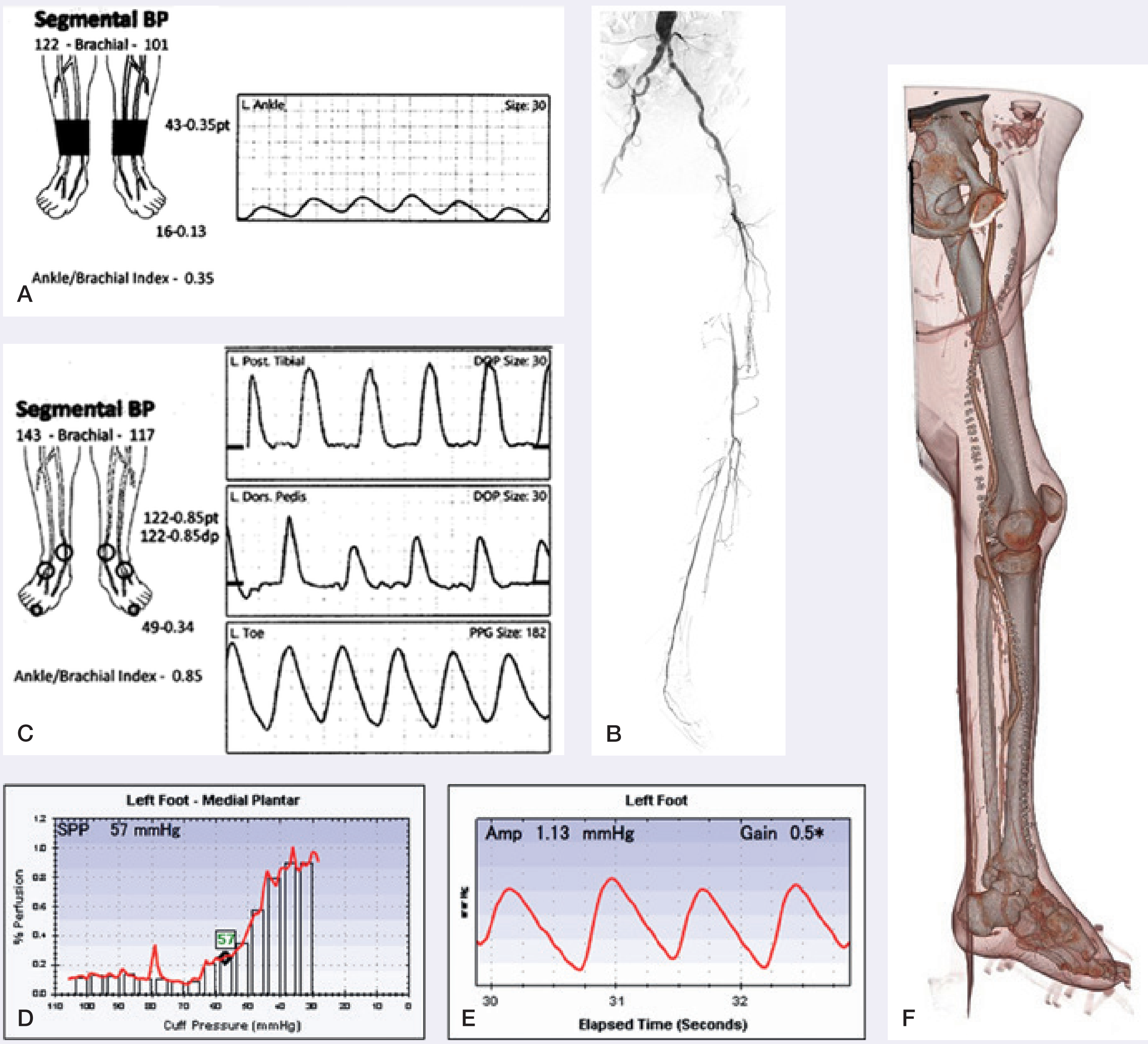


•AQ5•

FIGURE 3.8 Demonstration of an angiography and noninvasive studies without the need of arterial intervention.

CASE STUDY 3

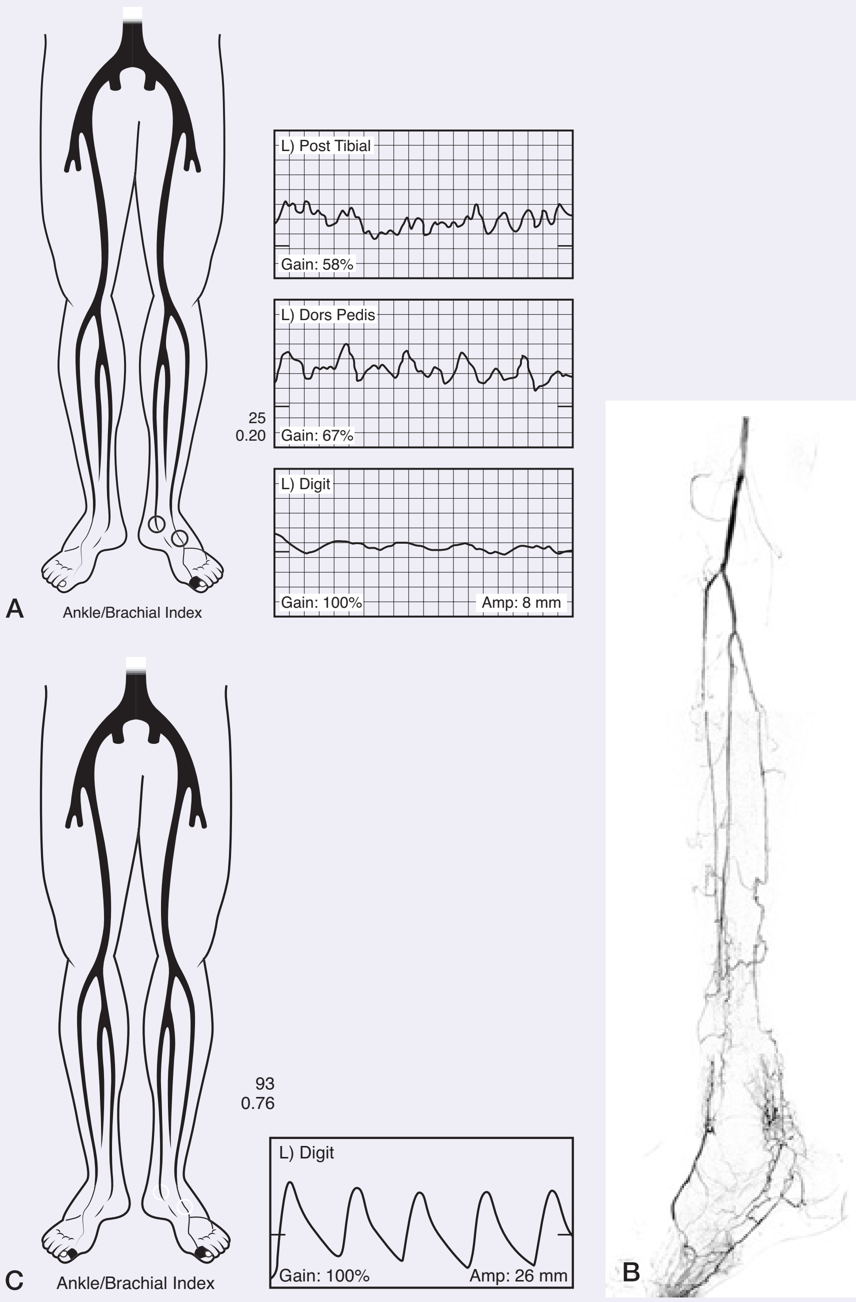
A 72-year-old man with diabetes mellitus (hemoglobin A_{1c} 7.6%) and peripheral neuropathy presented with a longstanding first toe ulcer and radiographic changes suggested of osteomyelitis in the first distal phalanx. Noninvasive vascular testing showed an ankle pressure of 43 mm Hg and a second toe pressure of 16 mm Hg. Plethysmography at the level of the ankle suggested severe impairment (Fig. 3.9A). Angiography showed occlusion of the superficial femoral artery from its origin through the level of the adductor hiatus, occlusion of the mid and distal anterior tibial artery and dorsalis pedis artery, and severe stenosis in the proximal posterior tibial artery (see Fig. 3.9B). A bypass from the common femoral to the midcalf posterior tibial artery was performed; because of a history of saphenectomy for previous coronary artery bypass grafting, this was done using a prosthetic vascular graft conduit and a distal vein patch constructed from the upper arm basilic vein. Repeat noninvasive testing following bypass demonstrated significant increases in ankle and toe pressures (122 and 49 mm Hg, respectively; see Fig. 3.9C). An SPP of 57 mm Hg also suggested adequate arterial perfusion (see Fig. 3.9D). Transmetatarsal plethysmography showed suggested only mild impairment (see Fig. 3.9E). Partial hallux amputation with primary wound closure was performed. Complete wound healing was seen. CTA done to evaluate for seroma demonstrated a widely patent bypass graft (see Fig. 3.9F).



•AQ5• **FIGURE 3.9** Demonstration of an arterial bypass from the common femoral to the midcalf posterior tibial artery by using a prosthetic vascular graft conduit and a distal vein patch constructed from the upper arm basilic vein.

CASE STUDY 4

An 85-year-old man with diabetes mellitus (hemoglobin A_{1c} of 6.8%) presented with a nonhealing wound and persistent cellulitis over the site of a fifth toe amputation done 1 month prior at another hospital. First toe pressure was 25 mm Hg, and ankle plethysmography showed severe impairment (Fig. 3.10A). Angiography showed inline flow from the aorta through the popliteal artery without hemodynamically significant lesions. Occlusions of all three tibial vessels were seen at the level of the mid to distal ankle with reconstitution of the dorsalis pedis artery, the pedal arch, and the lateral plantar artery (see Fig. 3.10B). An attempt at angioplasty of the distal anterior tibial artery was unsuccessful. A bypass from the popliteal artery to the dorsalis pedis artery using nonreversed ipsilateral saphenous vein was performed. Repeat first toe pressure following bypass was 93 mm Hg, and digit plethysmography was normal or mildly impaired (see Fig. 3.10C). Ostectomy of the fifth metatarsal head with primary wound closure was performed. The wound healed completely and cellulitis resolved.



•AQ5• **FIGURE 3.10** Demonstration of an arterial bypass from the popliteal artery to the dorsalis pedis artery using a nonreversed ipsilateral saphenous vein.

Conclusion

Noninvasive vascular testing can be very helpful in assuring that arterial blood flow to the foot is adequate for wound healing. Nonetheless, these are only numeric representations of a complex physiologic process, and limitations are inherent to these testing modalities. Clinicians should have a high index of suspicion in identifying peripheral artery disease. Palpable pedal pulses should be corroborated with noninvasive testing before significant PAD is ruled out. SPPs and toe pressures are probably the best noninvasive modalities for evaluating arterial perfusion in patients with diabetic foot ulcers. Anatomic imaging does not provide physiologic information and should therefore generally be reserved for preoperative planning.

References

1. Aragón-Sánchez J, Lázaro-Martínez JL. Factors associated with calcification in the pedal arteries in patients with diabetes and neuropathy admitted for foot disease and its clinical significance. Int J Low Extrem Wounds 2013;12:252–255.
2. Aro H, Eerola E, Aho AJ, et al. Tissue oxygen tension in externally stabilized tibial fractures in rabbits during normal healing and infection. J Surg Res 1984;37:202–207.
3. Attinger C, Cooper P, Blume P, et al. The safest surgical incisions and amputations applying the angiosome principles and using the Doppler to assess the arterial-arterial connections of the foot and ankle. Foot Ankle Clin 2001;6:745–799.
4. Attinger CE, Evans KK, Bulan E, et al. Angiosomes of the foot and ankle and clinical implications for limb salvage: reconstruction, incisions, and revascularization. Plast Reconstr Surg 2006;117(7 suppl):261–293.
5. Attinger CE, Meyr AJ, Fitzgerald S, et al. Preoperative Doppler assessment for transmetatarsal amputation. J Foot Ankle Surg 2010;49:101–105.
6. Barshes NR, Flores E, Belkin M, et al. The accuracy and cost-effectiveness of strategies used to identify peripheral artery disease among patients with diabetic foot ulcers. J Vasc Surg 2016;64:1682–1690.
7. Bianchi C, Montalvo V, Ou HW, et al. Pharmacologic risk factor treatment of peripheral arterial disease is lacking and requires vascular surgeon participation. Ann Vasc Surg 2007;21:163–166.
8. Blumenthal HT. More on microvascular disease of the foot in diabetes. N Engl J Med 1985;313:696–697.
9. Boyko EJ, Ahroni JH, Davignon D, et al. Diagnostic utility of the history and physical examination for peripheral vascular disease among patients with diabetes mellitus. J Clin Epidemiol 1997;50:659–668.
10. Boyko EJ, Ahroni JH, Stensel V, et al. A prospective study of risk factors for diabetic foot ulcer. The Seattle Diabetic Foot Study. Diabetes Care 1999;22:1036–1042.
11. Cormack GC, Lamberty BGH. The arterial anatomy of skin flaps. London, United Kingdom: Churchill Livingstone, 1989.
12. Demling RH. Nutrition, anabolism, and the wound healing process: an overview. Eplasty 2009;9:e9.
13. Diehm N, Rohrer S, Baumgartner I, et al. Distribution pattern of infrageniculate arterial obstructions in patients with diabetes mellitus and renal insufficiency—implications for revascularization. Vasa 2008;37:265–273.
14. Durrand JW, Batterham AM, O'Neill BR, et al. Prevalence and implications of a difference in systolic blood pressure between one arm and the other in vascular surgical patients. Anaesthesia 2013;68:1247–1252.
15. Ener BK. Dorsal pedal venous oximetry as an outcome index of lower extremity revascularizations. Vasc Surg 2001;35:37–41.
16. Espinola-Klein C, Rupprecht HJ, Bickel C, et al. Different calculations of ankle-brachial index and their impact on cardiovascular risk prediction. Circulation 2008;118:961–967.
17. Faglia E, Clerici G, Caminiti M, et al. Predictive values of transcutaneous oxygen tension for above-the-ankle amputation in diabetic patients with critical limb ischemia. Eur J Vasc Endovasc Surg 2007;33:731–736.
18. Faglia E, Favales F, Quarantiello A, et al. Angiographic evaluation of peripheral arterial occlusive disease and its role as a prognostic determinant for major amputation in diabetic subjects with foot ulcers. Diabetes Care 1998;21:625–630.
19. Fiordaliso F, Clerici G, Maggioni S, et al. Prospective study on microangiopathy in type 2 diabetic foot ulcer. Diabetologia 2016;59:1542–1548.
20. Fishbein MC, Fishbein GA. Arteriosclerosis: facts and fancy. Cardiovasc Pathol 2015;24:335–342.
21. Gray GH, Grant AA, Kalbaugh CA, et al. The impact of isolated tibial disease on outcomes in the critical limb ischemic population. Ann Vasc Surg 2010;24:349–359.
22. Hart CR, Layec G, Trinity JD, et al. Evidence of preserved oxidative capacity and oxygen delivery in the plantar flexor muscles with age. J Gerontol A Biol Sci Med Sci 2015;70:1067–1076.
23. Heinonen I, Saltin B, Kemppainen J, et al. Skeletal muscle blood flow and oxygen uptake at rest and during exercise in humans: a pet study with nitric oxide and cyclooxygenase inhibition. Am J Physiol Heart Circ Physiol. 2011;300:H1510–1517.
24. Huang CL, Wu YW, Hwang CL, et al. The application of infrared thermography in evaluation of patients at high risk for lower extremity peripheral arterial disease. J Vasc Surg 2011;54:1074–1080.
25. Jaszczak P. Skin oxygen tension, skin oxygen consumption, and skin blood flow measured by a tc-pO2 electrode. Acta Physiol Scand Suppl 1991;603:53–57.
26. Jawahar D, Rachamalla HR, Rafalowski A, et al. Pulse oximetry in the evaluation of peripheral vascular disease. Angiology 1997;48:721–724.
27. Khan NA, Rahim SA, Anand SS, et al. Does the clinical examination predict lower extremity peripheral arterial disease? JAMA 2006;295:536–546.
28. Lo T, Sample R, Moore P, et al. Prediction of wound healing outcome using skin perfusion pressure and transcutaneous oximetry: a single-center experience in 100 patients. Wounds 2009;21:310–316.
29. LoGerfo FW, Coffman JD. Current concepts. Vascular and microvascular disease of the foot in diabetes. Implications for foot care. N Engl J Med 1984;311:1615–1619.
30. Marston WA, Davies SW, Armstrong B, et al. Natural history of limbs with arterial insufficiency and chronic ulceration treated without revascularization. J Vasc Surg 2006;44:108–114.
31. Mills JL Sr, Conte MS, Armstrong DG, et al. The Society for Vascular Surgery Lower Extremity Threatened Limb Classification

System: risk stratification based on wound, ischemia, and foot infection (WIfI). J Vasc Surg 2014;59:220–234.e1.
32. Norgren L, Hiatt WR, Dormandy JA, et al. Inter-society consensus for the management of peripheral arterial disease (TASC II). J Vasc Surg 2007;45S:S5–67.
•AQ4• 33. Parameswaran GI, Brand K, Dolan J. Pulse oximetry as a potential screening tool for lower extremity arterial disease in asymptomatic patients with diabetes mellitus. Arch Intern Med 2005;165:442–446.
34. Pomposelli FB, Kansal N, Hamdan AD, et al. A decade of experience with dorsalis pedis artery bypass: analysis of outcome in more than 1000 cases. J Vasc Surg 2003;37:307–315.
35. Raffetto JD, Chen MN, LaMorte WW, et al. Factors that predict site of outflow target artery anastomosis in infrainguinal revascularization. J Vasc Surg 2002;35:1093–1099.
36. Roukis TS. The Doppler probe for planning septofasciocutaneous advancement flaps on the plantar aspect of the foot: anatomical study and clinical applications. J Foot Ankle Surg 2000;39:270–290.
37. Rueda CA, Nehler MR, Perry DJ, et al. Patterns of artery disease in 450 patients undergoing revascularization for critical limb ischemia: implications for clinical trial design. J Vasc Surg 2008;47:995–999.
38. Staffa E, Bernard V, Kubicek L, et al. Infrared thermography as option for evaluating the treatment effect of percutaneous transluminal angioplasty by patients with peripheral arterial disease. Vascular 2017;25;42–49.
39. Stoner HB, Barker P, Riding GS, et al. Relationships between skin temperature and perfusion in the arm and leg. Clin Physiol 1991;11(1):27–40.
40. Terasaki H, Inoue Y, Sugano N, et al. A quantitative method for evaluating local perfusion using indocyanine green fluorescence imaging. Ann Vasc Surg 2013;27:1154–1161.
41. Tsai FW, Tulsyan N, Jones DN, et al. Skin perfusion pressure of the foot is a good substitute for toe pressure in the assessment of limb ischemia. J Vasc Surg 2000;32:32–36.
42. Vierron E, Halimi JM, Giraudeau B. Ankle-brachial index and peripheral arterial disease. N Engl J Med 2010;362:471.
43. Weinberg I, Gona P, O'Donnell CJ, et al. The systolic blood pressure difference between arms and cardiovascular disease in the Framingham Heart Study. Am J Med 2014;127:209–215.
44. Yang C, Weng H, Chen L, et al. Transcutaneous oxygen pressure measurement in diabetic foot ulcers: mean values and cutpoint for wound healing. J Wound Ostomy Continence Nurs 2013;40:585–589.
45. Zheng J, Hasting MK, Zhang X, et al. A pilot study of regional perfusion and oxygenation in calf muscles of individuals with diabetes with a noninvasive measure. J Vasc Surg 2014;59:419–426.

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AQ3: Please confirm if use of "arteriography" is intended or should be changed to "angiography" per previous edition.

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